

Problem-Solution Fit Map

Smart-Lender Project Documentation Suite

Project Name	Smart Lender - AI-Powered Loan Approval Prediction System
Document Type	Problem-Solution Fit Map
Project Phase	Phase 3: Project Design Phase
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Status / Version	Approved / v1.0 (Exact Replica Format)

Problem-Solution Fit Grid Layout

Fit Parameter	Operational Challenges	Project Solution Response
Target Segment	Manual loan underwriters experiencing fatigue	Smart Lender interactive scoring dashboard
Technical Constraint	Inability to process textual elements	Integrated categorical encoders mapping options to floats
Risk Threshold	Biased predictions toward high approvals	Oversampling training labels using SMOTE before model execution
Time Constraints	Turnaround times stretching over days	Instant execution of mathematical inference logic in <0.5 seconds